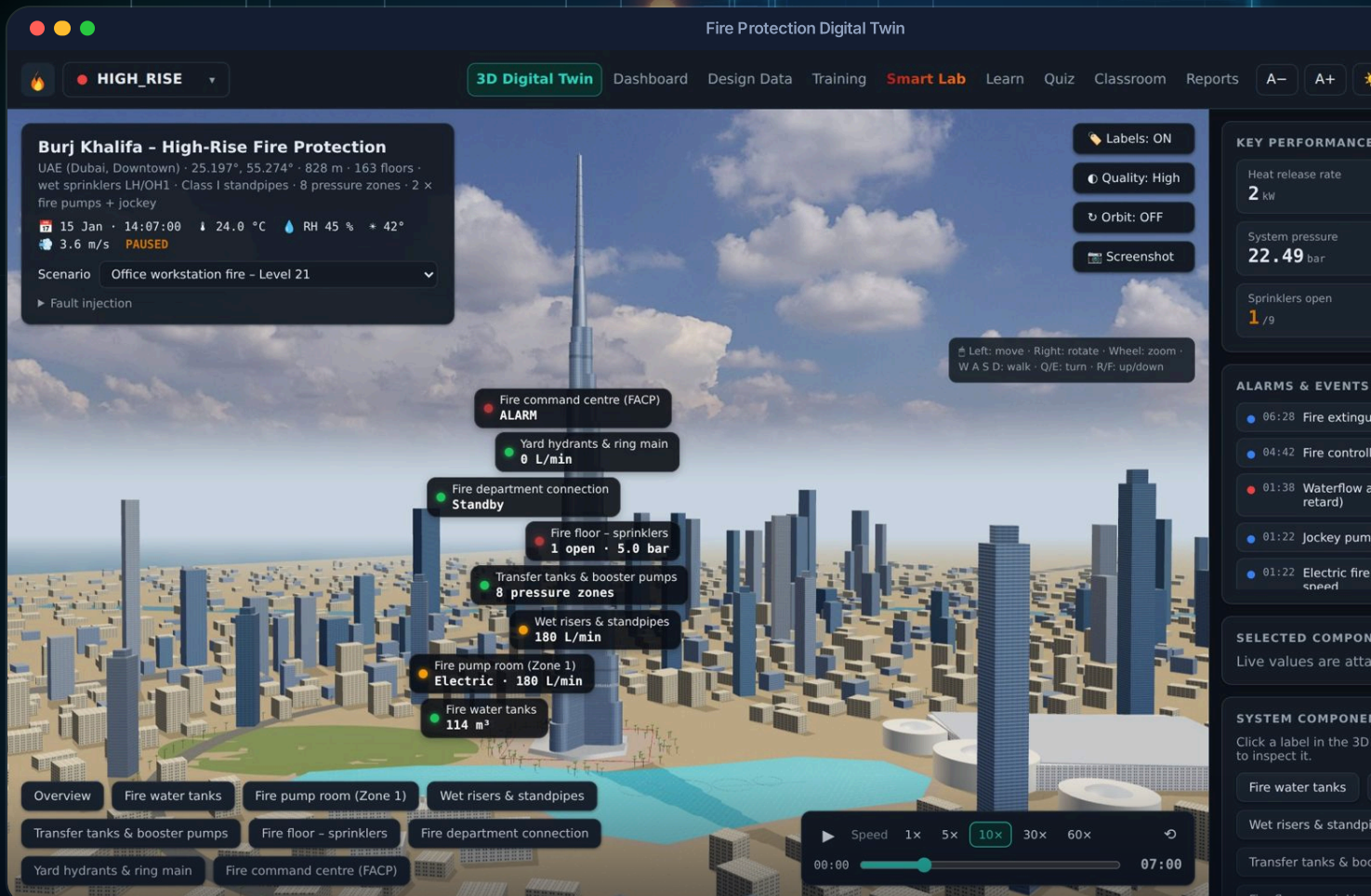


FIRE PROTECTION DIGITAL TWIN

Fire protection, alive in 3D.

Design, simulate, train and certify — sprinklers, pumps, smart fire alarm and special hazards — in one interactive digital twin.



5

Real reference facilities

9

Live fire scenarios

10

Hands-on 3D training scenes

10

Smart Systems Lab modules

14

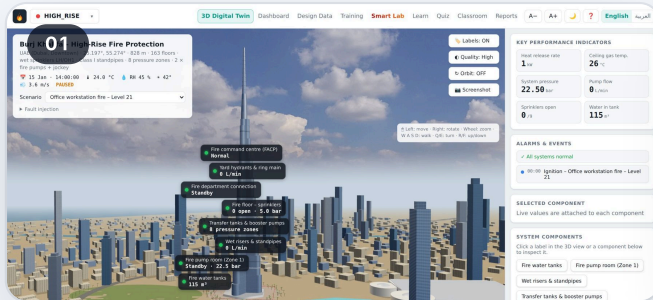
Illustrated lessons

130+

Exam-style questions

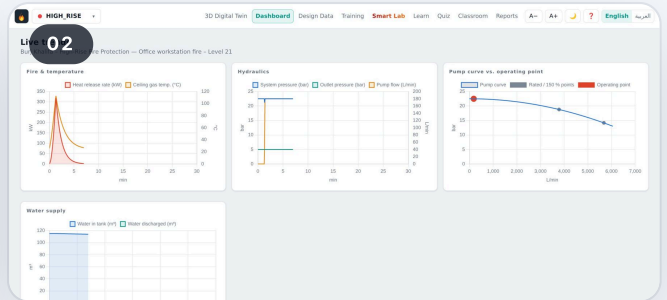
One app. Everything a fire engineer needs to see it work.

Instead of reading about a system, users walk through it, start it, break it and fix it — then prove what they learned.



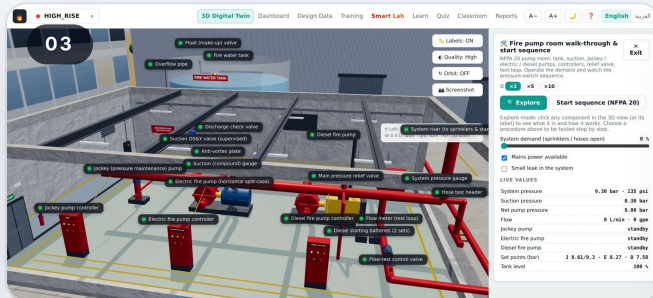
3D Digital Twin

Real buildings with every fire system modelled — run a fire and watch the system respond.



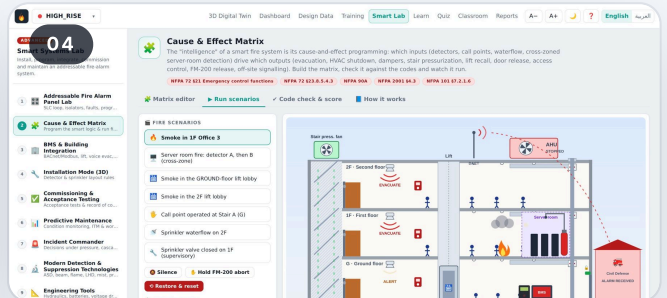
Live Dashboard

Fire growth, temperatures, pressures, pump curve and water supply — plotted live.



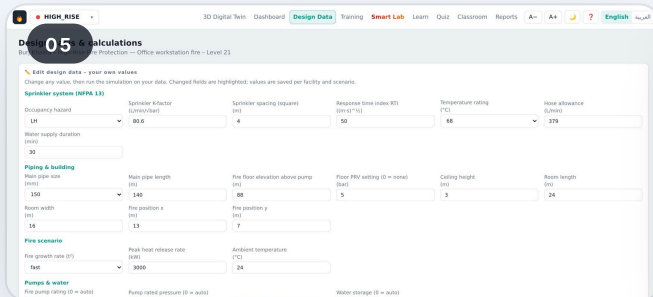
Hands-on 3D Training

Pump rooms, valves, extinguishers, FM-200 and installation — step-by-step and scored.



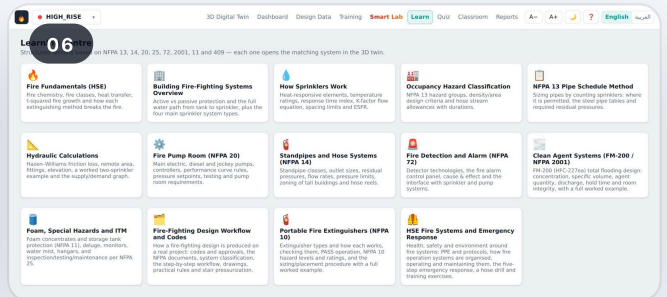
Smart Systems Lab

Addressable fire alarm, cause & effect, BMS integration, commissioning and maintenance.



Design Data & Tools

Edit design parameters and run engineering calculations — hydraulics, batteries, agents.



Learn · Quiz · Classroom

Lessons, exams, class management, PDF reports and certificates.

Windows 10 / 11 desktop app

Arabic & English · RTL

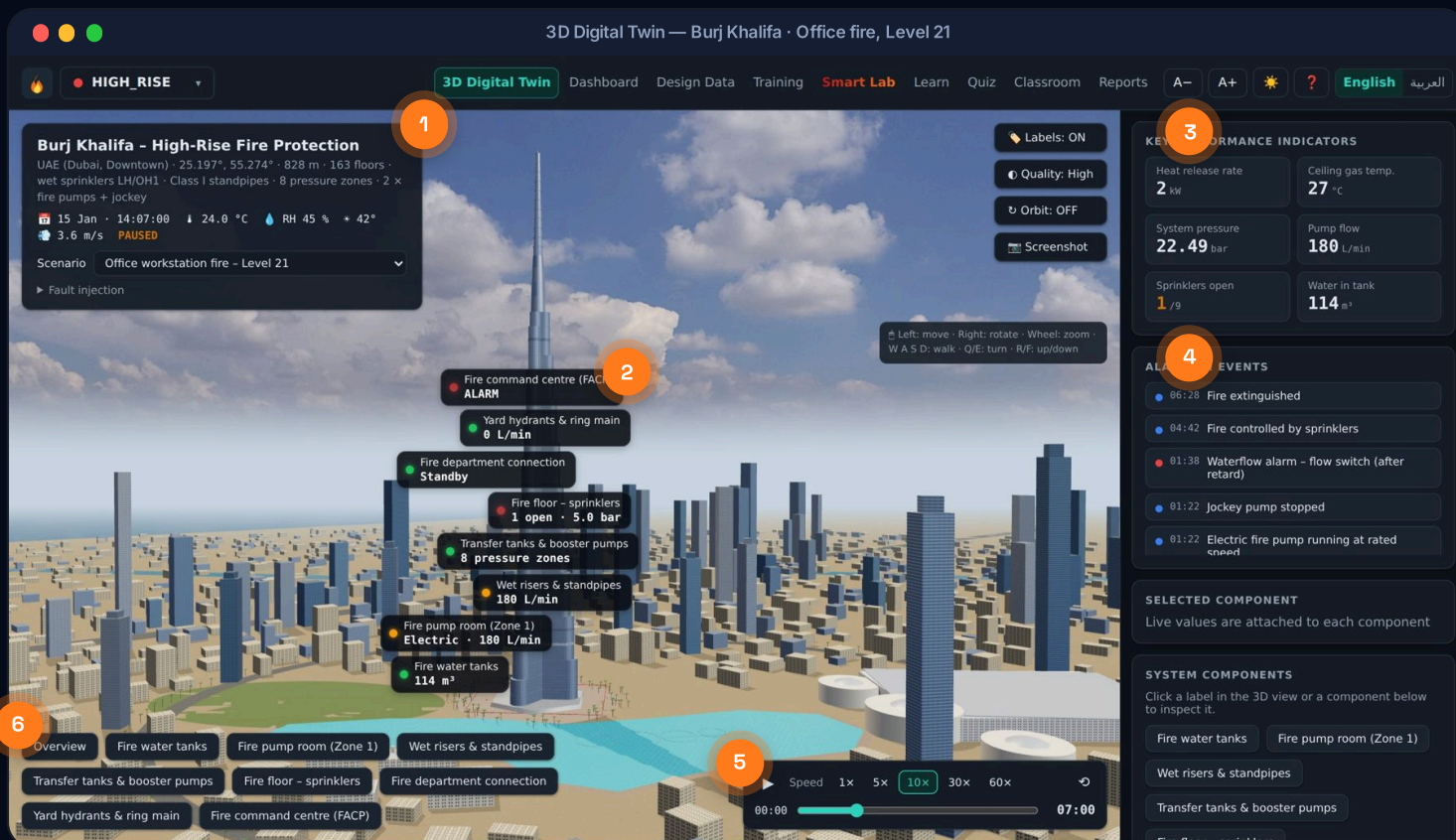
Works fully offline

Light & dark themes

Update notifications

Start a fire. Watch every system respond.

A time-driven simulation links the fire, the sprinklers, the pumps and the water supply. Pause it, speed it up, or inject a fault — and see the consequence instantly.



1 Scenarios & faults
Pick a fire scenario, then inject a fault — closed valve, power failure, low tank.

2 Live component labels
Each tank, pump, riser and floor shows its real-time state.

3 Key performance indicators
Heat release, gas temperature, pressure, pump flow, sprinklers open.

4 Alarms & events
A time-stamped log: ignition, waterflow, pump start, fire controlled.

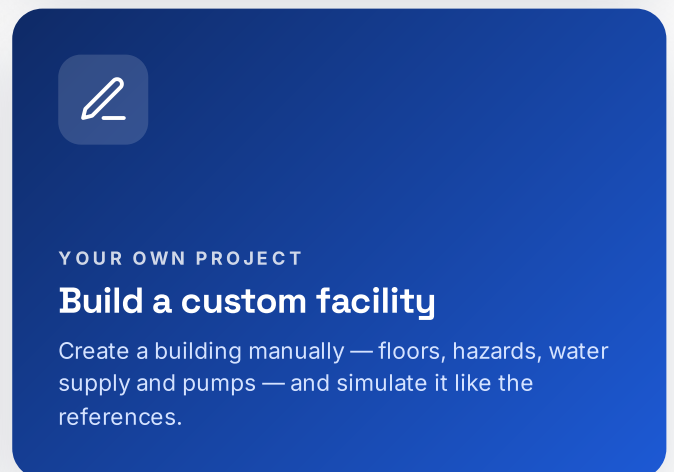
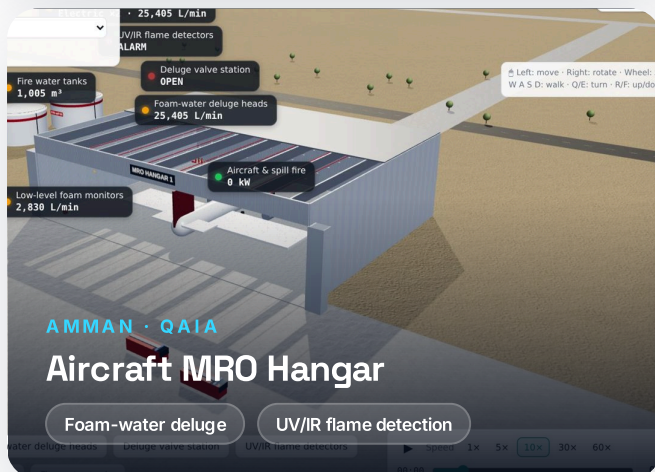
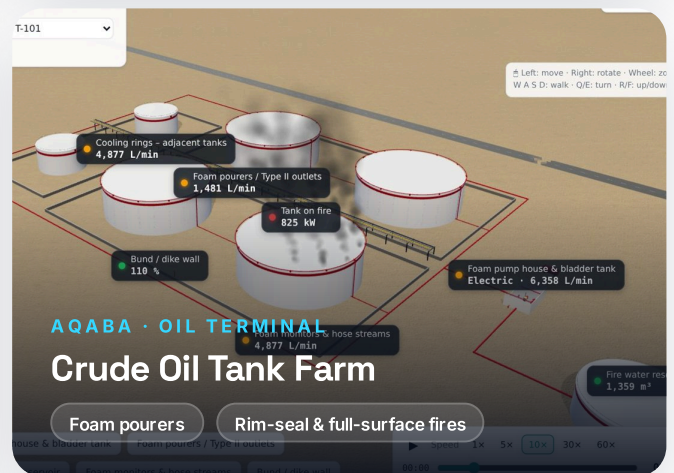
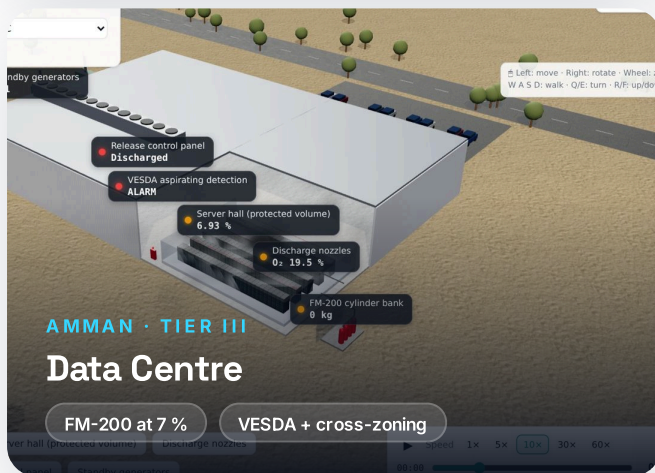
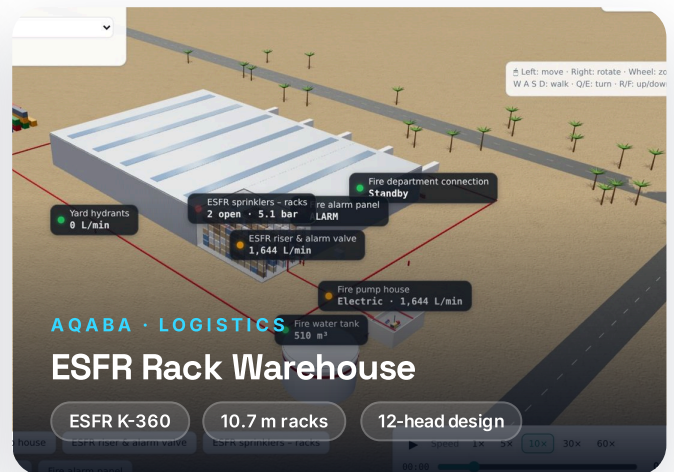
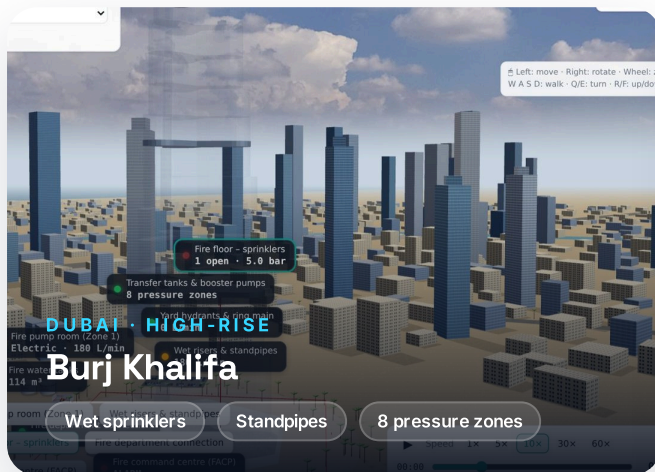
5 Time control
Play, pause and run at 1x to 60x — replay any moment.

6 Guided tour
One click focuses the view on each system: tanks, pumps, risers, floors.

Five real-world projects.

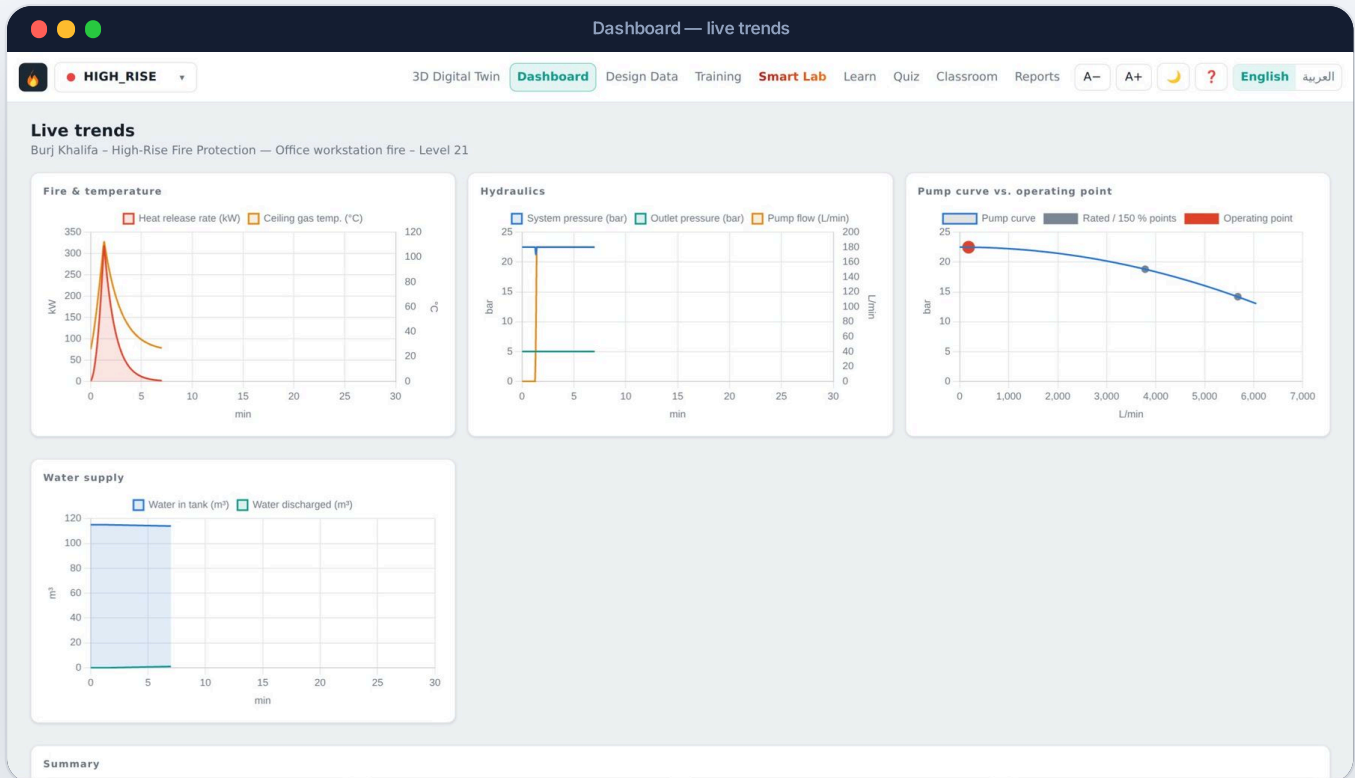
Five different hazards.

Each facility is a complete engineered model — water supply, pumps, risers, zones and detection — with its own fire scenarios.



The numbers behind every second of the fire.

Every simulation produces engineering-grade results — read them live, compare them with the design, and export them to a report.



HEAT RELEASE

MW live

SYSTEM PRESSURE

bar per zone

PUMP CURVE

Q-H vs. operating point

WATER IN TANK

m³ drawdown

Design Data

3D Digital Twin **Dashboard** Design Data Training **Smart Lab** Learn Quiz Classroom Reports A- A+ English العربية

Design basis & calculations
Burj Khalifa - High-Rise Fire Protection — Office workstation fire - Level 21

Edit design data - your own values
Change any value, then run the simulation on your data. Changed fields are highlighted; values are saved per facility and scenario.

Sprinkler system (NFPA 13)
Design hazard: 140
Sprinkler factor: 80.0
Sprinkler spacing (sqm): 4
Response time index RTI: 50
Temperature rating: 68
Rise allowance (L/min): 370
Water supply duration (min): 30

Piping & building
Main pipe size (mm): 150
Main pipe length (m): 140
Fire floor elevation above pump (m): 80
Floor FRI setting (0 = none) (bar): 5
Ceiling height (m): 3
Room length (m): 24
Room width (m): 14
Fire position x (m): 13
Fire position y (m): 7

Fire scenario
Peak heat release rate (kW): 3000
Fire growth rate (s⁻¹): 600
Ambient temperature (°C): 24

Pumps & water
Fire pump rating (0 = auto) (L/min): 600
Pump rated pressure (0 = auto) (bar): 0
Water storage (0 = auto) (m³): 0
Number of duty pumps (0 = auto): 0

Run simulation with my data **Recalculate** **Reset to reference design** **Export JSON** **Import JSON**



Editable design basis. Hazard class, density, design area, pipe sizes, pumps and tanks.



Hydraulics that react. Change a value and re-run — pressures and flows update.



One-click PDF reports of the scenario, timeline and results.

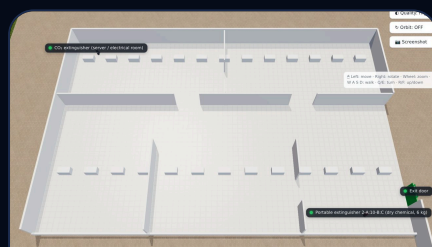
A screenshot of a virtual fire pump room environment. The room features a large red horizontal pipe system with various pumps and valves. Labels with green circular icons point to the following components:

- Discharge check valve
- Suction OS&Y valve (supervised)
- Anti-vortex plate
- Jockey (pressure maintenance) pump
- Electric fire pump (horizontal split-case)
- Main pressure relief valve
- System pressure gauge
- Hose test header
- Jockey pump controller
- Electric fire pump controller
- Diesel fire pump controller
- Flow meter (test loop)
- Diesel starting batteries (2 sets)
- Diesel fire pump
- System riser (to sprinklers & WA S Drains) (100mm K1 Support)

Jockey, electric and diesel pumps, controllers and pressure set-points (NFPA 20).



Trip, drain and reset.



Pick the right class, then put it out.



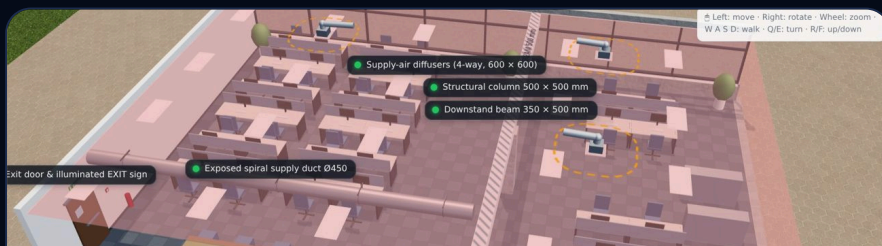
Types, temperatures & RTI.



Smoke control per NFPA 92.



Agent calculator & discharge.



Place detectors and sprinklers on a real plan — spacing and coverage are checked against the code.



Smart Systems Lab — Cause & Effect run

HIGH_RISE
▼

3D Digital Twin
Dashboard
Design Data
Training

Smart Lab

Learn
Quiz
Classroom
Reports

A-

A+

🌙

?

English

العربية

ADVANCED

Smart Systems Lab

Install, program, integrate, commission and maintain an addressable fire-alarm system.

- 1

Addressable Fire Alarm Panel Lab

SLC loop, isolators, faults, progr...
- 2

Cause & Effect Matrix

Program the smart logic & run f...
- 3

BMS & Building Integration

BACnet/Modbus, lift, voice evac...
- 4

Installation Mode (3D)

Detector & sprinkler layout rules
- 5

Commissioning & Acceptance Testing

Acceptance tests & record of co...
- 6

Predictive Maintenance

Condition monitoring, ITH & wor...
- 7

Incident Commander

Decisions under pressure, casca...
- 8

Modern Detection & Suppression Technologies

ASD, beam, flame, LHD, mist, pr...
- 9

Engineering Tools

Hydraulics, batteries, voltage dr...
- 10

Learning Paths & Certificates

Levels, final exam, certificates

Cause & Effect Matrix

The "intelligence" of a smart fire system is its cause-and-effect programming: which inputs (detectors, call points, waterflow, cross-zoned server-room detection) drive which outputs (evacuation, HVAC shutdown, dampers, stair pressurization, lift recall, door release, access control, FM-200 release, off-site signalling). Build the matrix, check it against the codes and watch it run.

NFPA 72 §21 Emergency control functions
NFPA 72 §23.8.5.4.3
NFPA 90A
NFPA 2001 §4.3
NFPA 101 §7.2.1.6

Matrix editor

Run scenarios

Code check & score

How it works

FIRE SCENARIOS

Smoke in 1F Office 3

Server room fire: detector A, then B (cross-zone)

Smoke in the GROUND-floor lift lobby

Smoke in the 2F lift lobby

Call point operated at Stair A (G)

Sprinkler waterflow on 2F

Sprinkler valve closed on 1F (supervisory)

Silence

Hold FM-200 abort

Restore & reset

OUTPUT STATUS

Horn/strobes off

Voice evacuation
G:alert 1:evac 2:evac

AHUs
G:stop 1:stop 2:stop

Smoke dampers
closed

Smoke pressurization
uniform

Stair press. fan

Lift

AHU STOPPED

2F - Second floor

1F - First floor

G - Ground floor

EVACUATE

EVACUATE

ALERT

[illegible]

NFPA 14

From first device to final certificate.

01



Addressable panel

Program devices, loops & zones.

02



Cause & effect

Build and test the matrix.

03



BMS integration

BACnet, Modbus, Contact-ID.

04



Installation rules

Spacing, wiring, mounting.

05



Commissioning

Acceptance tests & records.

06



Predictive maintenance

Trends before failures.

07



Incident commander

Decide under pressure.

08



Modern technologies

VESDA, beam, flame, mist.

09



Engineering tools

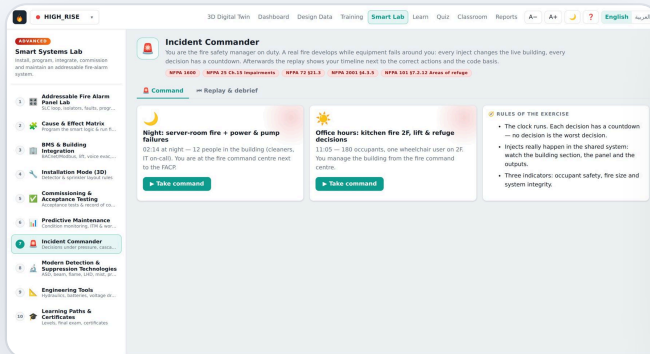
Batteries, hydraulics, agents.

10

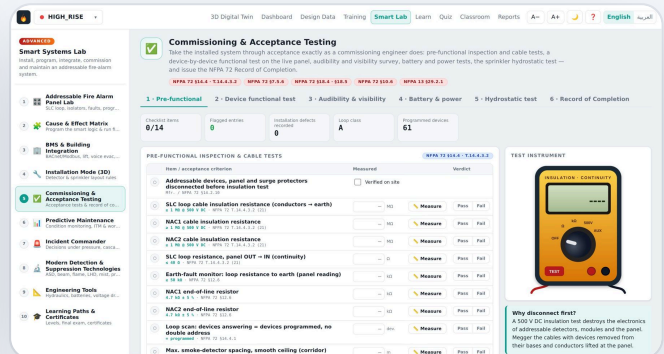


Paths & certificates

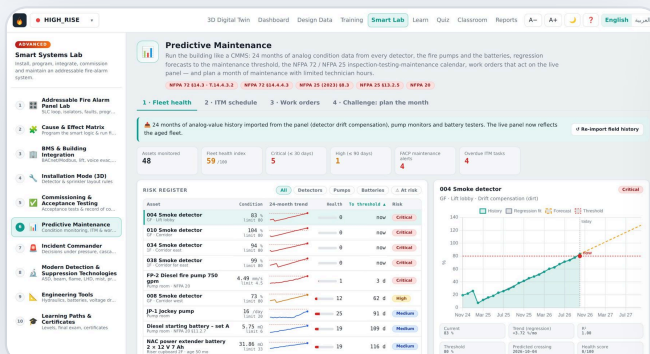
Levels, final exam, certificate.



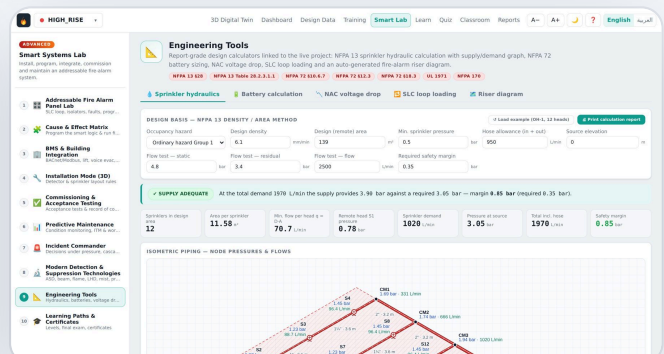
Incident commander · live decisions against the clock



Commissioning · device-by-device acceptance



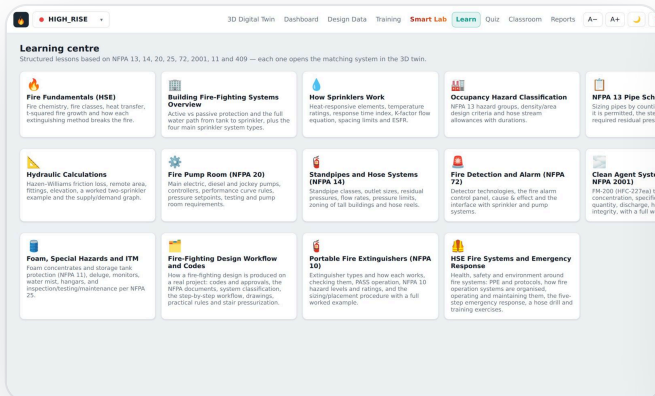
Predictive maintenance · detector drift & battery health



Engineering tools · calculators with code references

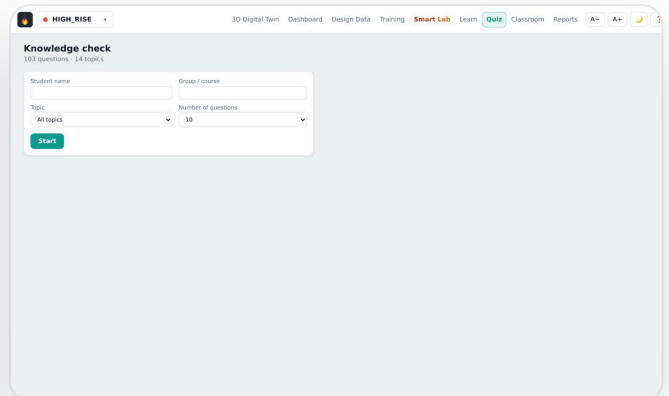
A complete course, built into the simulator.

Theory, practice and assessment in the same place — so instructors can run a full programme without extra tools.



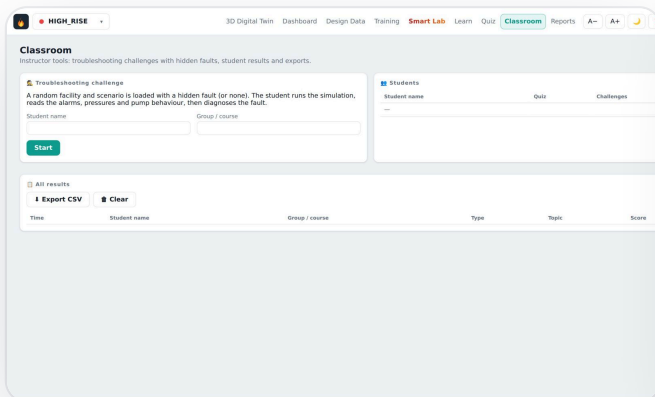
14 illustrated lessons

From fire basics and hazard classification to hydraulic calculations, clean agents and foam.



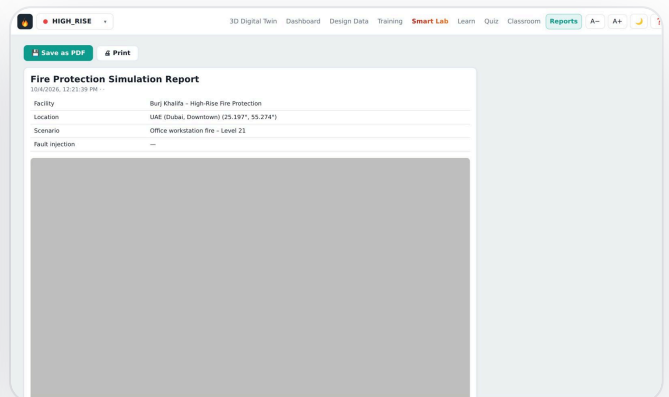
130+ exam-style questions

Instant feedback with the explanation and the code reference for every answer.



Classroom

Instructor tools: troubleshooting challenges with hidden faults, student results and CSV export.



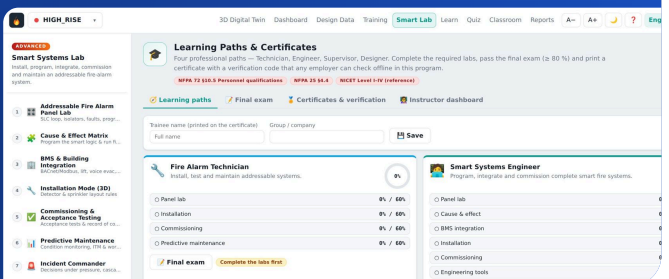
PDF reports

Professional scenario and trainee reports — ready to file, print or send.

LEARNING PATHS

Earn a verifiable certificate

Structured levels lead to a final exam; passing trainees receive a printable certificate with a unique verification code.



Built for the people who design, install and protect.



Universities

- Fire & MEP engineering courses
- Virtual lab for every student
- Graduation projects on real buildings



Training centres

- Ready-made programmes & exams
- Hidden-fault challenges & results
- Certificates for each level



Contractors & consultants

- Onboard engineers and technicians
- Explain designs to clients in 3D
- Commissioning & ITM practice



Facility, HSE & civil defence

- Emergency-response drills
- Operator awareness on their systems
- Incident-command decision training



Zero risk

Practise failures that can never be staged on a live system.



Faster competence

See cause and effect instead of imagining it.



Measured skills

Every scene and exam is scored and recorded.



Arabic & English

Teach in the language of your trainees.

THE IDEA

A complete fire-protection lab that fits on every laptop.

No pump room, no live panel, no shutdown permit — trainees can start, break and fix real systems as often as they need.

Take the whole twin — or just the part you teach.

Each license is tailored to the customer: the full program or a focused package, on a monthly or yearly subscription.

 Full program Every facility, scene, lab module and feature. MOST COMPLETE	 Sprinklers & pumps Water-based systems: sprinklers, pumps, valves and hydraulics. PACKAGE	 Smart fire alarm Addressable panels, cause & effect, BMS and commissioning. PACKAGE
 Special hazards FM-200, foam and aviation facilities. PACKAGE	 Basic awareness For schools and companies: extinguishers, hoses and lessons. PACKAGE	 Custom Pick exactly the facilities, scenes, lab modules and features you need. TAILORED
Monthly Ideal for a single course or a short programme.	Yearly Best value for universities and training centres.	14-day trial Every installation starts with a free full trial.
 PLATFORM Windows 10 / 11 · 64-bit	 MEMORY 8 GB RAM (16 GB recommended)	 GRAPHICS DirectX 11 / WebGL 2
 CONNECTIVITY Runs fully offline	 ACTIVATION Signed license per machine	 UPDATES In-app new-version notice



GET STARTED

See your trainees master fire protection.

Book a live demo and we will walk you through the twin on the systems you teach or operate.

[Book a live demo →](#)

1

Live demo

Online or on-site, in Arabic or English.

2

Free trial

14 days with the full program.

3

Your package

Full or focused · monthly or yearly.

4

Go live

Install, activate and start training.

CONTACT ASFAN

Let's build your training programme.



EMAIL

info@asfanco.com



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